Gap versus trial overlap: measured and certified filter cost on J_1–J_2 chains

Data report for the filter-pipeline paper

October 29, 2026

This report separates the two inputs that set the filter's cost, the spectral gap Δ and the trial overlap $\gamma = |\langle E_0|\psi\rangle|^2$. Each is held fixed across system size while the other is varied. Every *measured* number is an exact state-vector simulation of the Trotterised filter circuit. Every *certified* number is the pipeline's a-priori bound (Theorem 1) with the same inputs. Nothing is extrapolated. All tables and figures are generated by `sbpipe_study/gapov_report/make_report_gapov.py` from `results/gapov_tune.json` and `results/gapov.jsonl` (222 cases), produced by `sbpipe_study/run_gapov.py`.

1 Holding the gap and the overlap fixed

Model. Open J_1–J_2 Heisenberg chains, $J_1 = 1$, $N = 8, \ldots, 14$ qubits. $\Delta = E_1 - E_0$ is the singlet–triplet gap from exact diagonalisation.

Gap tuning. At each N, J_2 is found by bisection so that Δ hits the target to 10^{-9} (Table 1). The gap closes with N at fixed J_2, so larger chains need larger J_2. Some (gap, N) pairs have no solution below the dimerised regime and are left out.

Table 1: J_2 that gives each target gap Δ at each N ("–": not reachable).

Δ	$N = 8$	$N = 10$	$N = 12$	$N = 14$
0.30	–	–	0.4114	0.4636
0.35	–	0.4276	0.4911	0.5135
0.40	0.3838	0.5101	0.5387	0.5501
0.45	0.5109	0.5611	0.5755	0.5806
0.50	0.5762	0.6011	0.6071	0.6083
0.55	0.6252	0.6359	0.6405	0.6776

Overlap tuning with a real circuit. The trial is the variational $\chi = 16$ MPS, prepared by its staircase circuit, followed by $R_y(\theta) = e^{-i\theta Y/2}$ on every even site. A uniform rotation on all sites would do nothing, because the ground state is a singlet and so is invariant under global spin rotations. Rotating one sublattice breaks that symmetry and lowers γ smoothly from the MPS value. θ is found by bisection so that γ hits the target to 10^{-12} (Table 2). The trial cost is the MPS circuit plus $N/2$ single-qubit rotations, and is the same at every γ for a given N.

Grid. Study A fixes $\Delta \in \{0.40, 0.50\}$ and varies $\gamma \in \{0.03, 0.1, 0.3, 0.5, 0.7, 0.9\}$. Study B fixes $\gamma \in \{0.1, 0.5\}$ and varies Δ over 0.30–0.55. Both run at $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, for 222 cases in total.

2 How each cost is obtained

The cost is the expected number of CNOTs per successful (post-selected) run that leaves the state with infidelity $1 - F \le \varepsilon$:

$$C = \frac{C_{\text{filter}} + C_{\text{trial}}}{p}, \tag{1}$$

Table 2: Rotation angle θ that sets each overlap γ (shown at $\Delta = 0.40$).

γ	$N = 8$	$N = 10$	$N = 12$	$N = 14$
0.03	1.516	1.604	1.549	1.474
0.1	1.280	1.325	1.273	1.208
0.3	0.959	0.975	0.932	0.883
0.5	0.739	0.746	0.712	0.673
0.7	0.536	0.538	0.512	0.484
0.9	0.293	0.294	0.279	0.264
γ of $\chi = 16$ MPS ($\theta = 0$)	1.0000	1.0000	1.0000	1.0000

where p is the success probability ($p \approx \gamma$).

Measured, dyadic SBC. Spacing $\tau = \pi/(1 + \delta)$, where $\delta = \Delta/W$ is the gap in units of the spectral width W. The exact filter (matrix exponentials) is applied to the trial for each ladder (K, r). The two shortest ladders that reach ε are kept. Each is then Trotterised with second-order Strang steps, and the coarsest common step dt at which the full circuit still reaches ε is found. The cheaper of the two is reported.

Measured, QETU (reference). The smallest even Chebyshev degree whose exact filter reaches ε is found. The search then looks for the fewest Strang steps per query, or a slightly higher degree at one step, that still reaches ε after Trotterisation.

Certified. The pipeline's `refine()` with exact energies (E_0, E_1 errors 10^{-9}), the Weyl upper bound on the spectrum and the true γ. This gives the circuit that Theorem 1 guarantees reaches ε with commutator Trotter bounds, without simulating it.

3 Study A: gap fixed, overlap varied

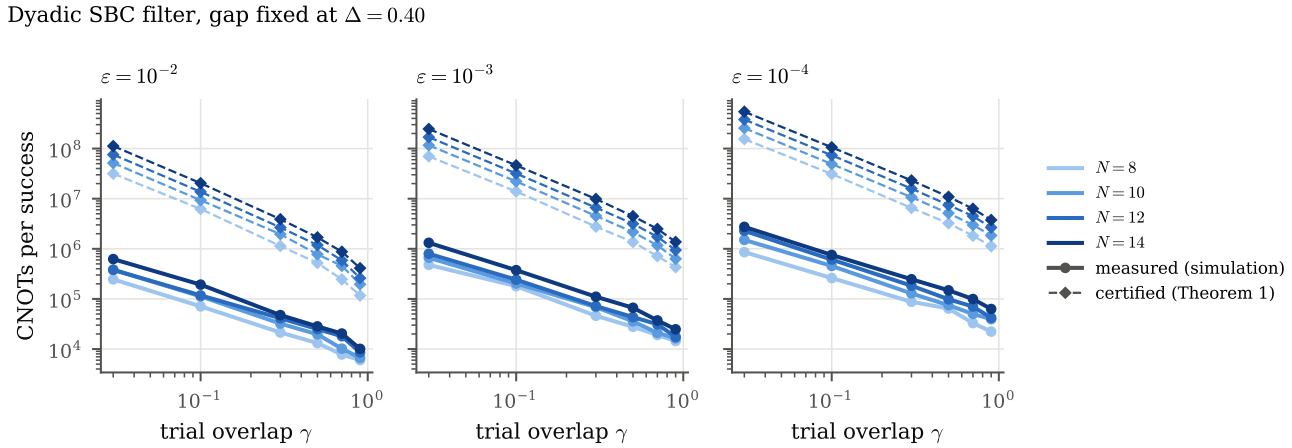


Figure 1: Dyadic SBC filter at $\Delta = 0.40$ for every N. Solid: measured by simulation. Dashed: certified.

Result. Overlap is the dominant input. At $\Delta = 0.40$, lowering γ from 0.9 to 0.03 (a factor of 30) raises the measured dyadic cost by 33–62× and the certified cost by 137–273×. Most of the rise is the repetition $1/p \approx 1/\gamma$: each success needs about $1/\gamma$ attempts. The rest comes from the longer filter needed to remove more excited weight (larger L). The curves for different N run nearly parallel, so the effect of the overlap does not depend on size over this range. Across all cases the certificate lies 19–290× (median 75×) above the measured dyadic cost. The excess is largest at small γ, because the bound grows faster in L than the measured cost does (Table 4).

Table 3: Dyadic filter at $\Delta = 0.40$, $\varepsilon = 10^{-3}$: measured / certified CNOTs per success.

γ	$N = 8$	$N = 10$	$N = 12$	$N = 14$
0.03	4.8×10^5 / 7.0×10^7	6.6×10^5 / 1.2×10^8	8.0×10^5 / 1.7×10^8	1.3×10^6 / 2.5×10^8
0.1	1.8×10^5 / 1.4×10^7	2.0×10^5 / 2.2×10^7	2.4×10^5 / 3.2×10^7	3.7×10^5 / 4.6×10^7
0.3	4.6×10^4 / 2.8×10^6	6.9×10^4 / 4.5×10^6	7.2×10^4 / 6.6×10^6	1.1×10^5 / 9.9×10^6
0.5	2.8×10^4 / 1.4×10^6	3.6×10^4 / 2.2×10^6	4.3×10^4 / 3.2×10^6	6.7×10^4 / 4.5×10^6
0.7	1.9×10^4 / 7.1×10^5	2.1×10^4 / 1.2×10^6	3.1×10^4 / 1.7×10^6	3.7×10^4 / 2.5×10^6
0.9	1.4×10^4 / 4.3×10^5	1.6×10^4 / 6.3×10^5	1.7×10^4 / 9.4×10^5	2.5×10^4 / 1.4×10^6

4 Study B: overlap fixed, gap varied

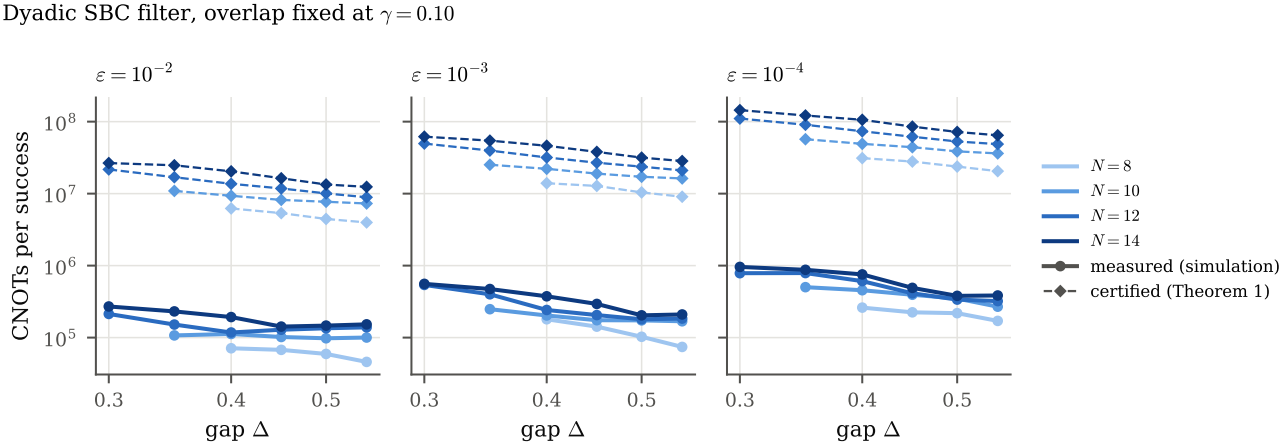


Figure 2: Dyadic SBC filter at $\gamma = 0.10$ with the gap held fixed across N. Solid: measured. Dashed: certified.

Result. At fixed overlap, the gap matters much less over the range that can be held fixed at every N. At $\gamma = 0.1$, closing Δ from 0.55 to 0.30 (a factor of 1.8) raises the measured dyadic cost by 1.1–2.9×, QETU by 1.4–1.8× and the certified cost by 1.5–2.4×. At fixed $\Delta = 0.40$ and $\gamma = 0.1$, growing the chain from $N = 8$ to 14 raises the measured cost 2.1–2.9× and the certified cost 3.3–3.4×. A wider gap range is not reachable at every N (Table 1), so the gap exponent is less well pinned than the overlap one.

5 Cost as a function of N, Δ, γ and ε

Every case, from both studies and all three ε, is fitted to one form:

$$C \;\approx\; a\, N^{\alpha}\, \Delta^{-\beta}\, L^{\lambda}\, \gamma^{-g}, \qquad L = \ln \frac{1-\gamma}{\gamma\,\varepsilon}. \tag{2}$$

L is the log factor every filter needs: to bring infidelity from $(1-\gamma)/\gamma$ down to ε, it must suppress the excited weight by $\eta^2 = \gamma\varepsilon/(1-\gamma)$.

The cost rules. With $L = \ln[(1-\gamma)/(\gamma\varepsilon)]$, the expected CNOTs per success are

$$C^{\text{meas}}_{\text{dyadic}} \approx 1.03 \times 10^1\; N^{1.42}\, \Delta^{-1.36}\, L^{1.71}\, \gamma^{-0.74} \qquad \text{(within 1.26 × typical, 2.1 × worst)}, \tag{3}$$

$$C^{\text{meas}}_{\text{QETU}} \approx 1.61 \times 10^1\; N^{2.13}\, \Delta^{-0.83}\, L^{0.78}\, \gamma^{-0.99} \qquad (1.04\times,\ 1.2\times), \tag{4}$$

$$C^{\text{cert}}_{\text{dyadic}} \approx 3.38 \times 10^1\; N^{2.03}\, \Delta^{-1.27}\, L^{2.44}\, \gamma^{-0.98} \qquad (1.22\times,\ 2.2\times), \tag{5}$$

$$C^{\text{cert}}_{\text{QETU}} \approx 2.01 \times 10^1\; N^{1.98}\, \Delta^{-1.12}\, L^{2.20}\, \gamma^{-0.98} \qquad (1.22\times,\ 1.9\times). \tag{6}$$

Table 4: Fitted exponents of Eq. (2), using every case. "rms" and "max" are the typical and worst factor by which a case differs from the fit.

	a	α (N)	β (Δ)	λ (L)	g (γ)	rms	max	cases
dyadic, measured	1.03×10^1	1.42	1.36	1.71	0.74	1.26×	2.1×	222
QETU, measured	1.61×10^1	2.13	0.83	0.78	0.99	1.04×	1.2×	222
dyadic, certified	3.38×10^1	2.03	1.27	2.44	0.98	1.22×	2.2×	222
QETU, certified	2.01×10^1	1.98	1.12	2.20	0.98	1.22×	1.9×	222

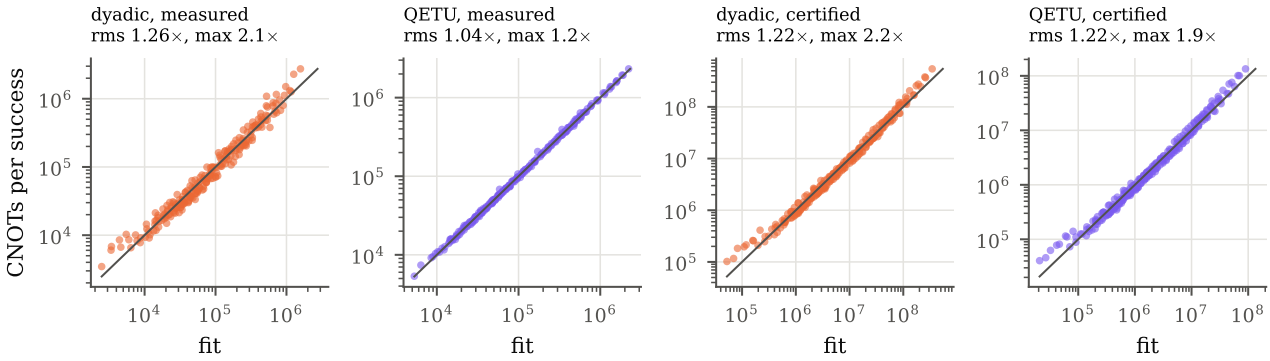


Figure 3: Each case against its fit, Eq. (2) with Table 4.

Three of the four rules have γ^{-1} almost exactly, which is the post-selection repetition. The measured dyadic rule has $\gamma^{-0.74}$, because its success probability ranges from 0.76 to 1.24 times γ: the ladders over-suppress, so part of its γ dependence is absorbed into L. The certified rules grow as $L^{2.2\text{–}2.4}$, where the measured ones grow as $L^{0.8\text{–}1.7}$. That difference is why the certificate's overhead grows at small γ and small ε. The certified N^2 is one factor of N from the CNOTs per Trotter step and one from the spectral width $W \propto N$, which sets the filter length at a fixed gap.

Dyadic versus QETU, measured. Over all cases, QETU costs a median 1.31× the dyadic filter. The dyadic filter is cheaper in every case at $\varepsilon = 10^{-2}$ and in all but 9 at 10^{-3}. At $\varepsilon = 10^{-4}$ QETU is cheaper in 56 of 74 cases, because the dyadic ladder lengthens in coarse power-of-two steps. The fits show the same thing: the dyadic cost grows as $L^{1.71}$ against $L^{0.78}$ for QETU, while the dyadic cost grows more slowly with N ($N^{1.42}$ against $N^{2.13}$).

6 Caveats

- The sizes span $N = 8$–14 (less than a factor of two), and the gaps span 0.30–0.55. The exponents summarise these data. They are not asymptotic laws, and the N exponent in particular is loose.
- The N exponent mixes two effects: the CNOTs per Trotter step (about $\propto N$) and the spectral width $W \approx 0.7N$, which sets the filter length at a fixed gap in absolute units.
- The certified fit summarises the bound over this grid. The bound itself (Theorem 1) holds case by case and does not depend on the fit.
- The dyadic filter is discrete: (K, r) and dt move in jumps, so its measured cost scatters more about the fit than QETU's does.